

(b) Density and pressure**Students should:**

5.3 know and use the relationship between density, mass and volume:

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

$$\rho = \frac{m}{V}$$

5.4 *practical: investigate density using direct measurements of mass and volume*

5.5 know and use the relationship between pressure, force and area:

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

$$p = \frac{F}{A}$$

5.6 understand how the pressure at a point in a gas or liquid at rest acts equally in all directions

5.7 know and use the relationship for pressure difference:

pressure difference = height × density × gravitational field strength

$$p = h \times \rho \times g$$